

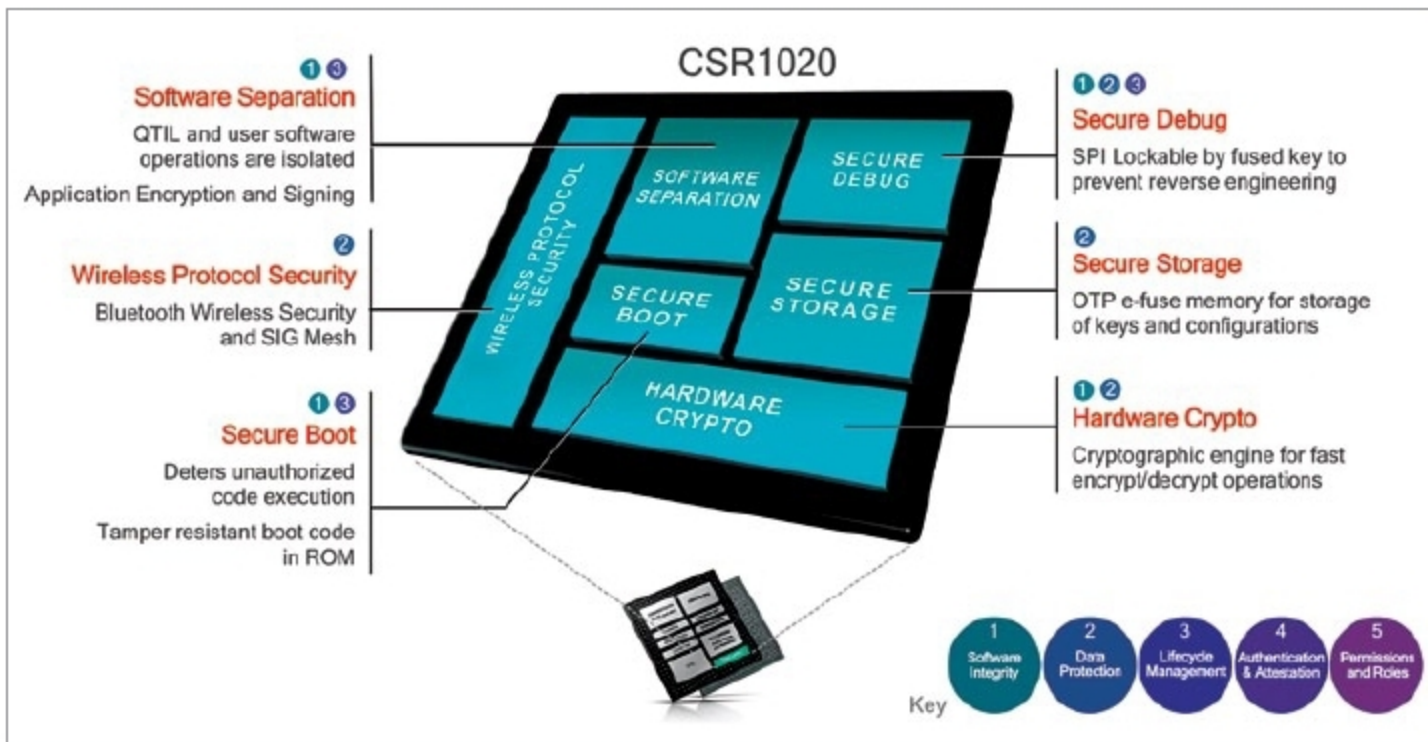


Fig. 2: Bluetooth SoC security foundations (Credit: Qualcomm Technologies)

Hwai Lin Khor, head of sales and business development - chipcard and security, Infineon Technologies Asia Pacific Pte Ltd, says, "Main components of a device include MCU, non-volatile memory (optionally embedded in MCU) and security IC that implements hardware-based security capabilities. Sensors and actuators complete the design. The key is to map the secure software update process to the device's architecture.

"Infineon's OPTIGA Trust X meets these security challenges faced by manufacturers. It is a turnkey, hardware-based solution offering enhanced security for connected devices, including secure firmware update functionality. It is easy to integrate and is TCG-compliant."

Software-based security

For connected devices that can be controlled remotely, it is essential to have the ability to install new software. Software updates are used to add new features, fix bugs and resolve known security weaknesses. Highly automated software update processes that require little interaction and reduce downtime, however, represent pos-

sible entry points for attackers. If the connected devices are not properly protected, these may be left open to manipulation, typically through the installation of a malicious code on the devices, allowing the attacker to do what he or she wants it to do, which can include taking control of critical infrastructures, industrial automation systems or even cars.

With many of today's connected devices, the user has to provide a password before the software image can be installed. The software itself may then have an additional layer of pro-

tection for integrity, for example, a cryptographic checksum (hash). The password protection scheme can be replaced or enhanced by cryptographic mechanisms, enabling authenticity and integrity checks, policy enforcement processing as well as optional confidentiality protection.

Software update is a set of data that contains the software code and information required for its proper installation. It contains the meta-data describing software update image and conditions.

Metadata of a software update data set contains a wide range of data, including image version, image hash, image reference data and information on how the connected device should deal with the software update image.

Steps performed during a software update are:

1. Secure storage of public key and secure verification of metadata signature

2. Secure storage and handling of update master key for decrypting software image

3. Copying software update image to target location

4. Checking integrity of software image

5. Activating new software image

Without further security measures, these data fields can be manipulated by an attacker. To equip the metadata with capabilities for securing the update process, following cryptographic steps are required:

1. Metadata itself must be integrity-protected. This is typically done via a cryptographic checksum called a hash value.

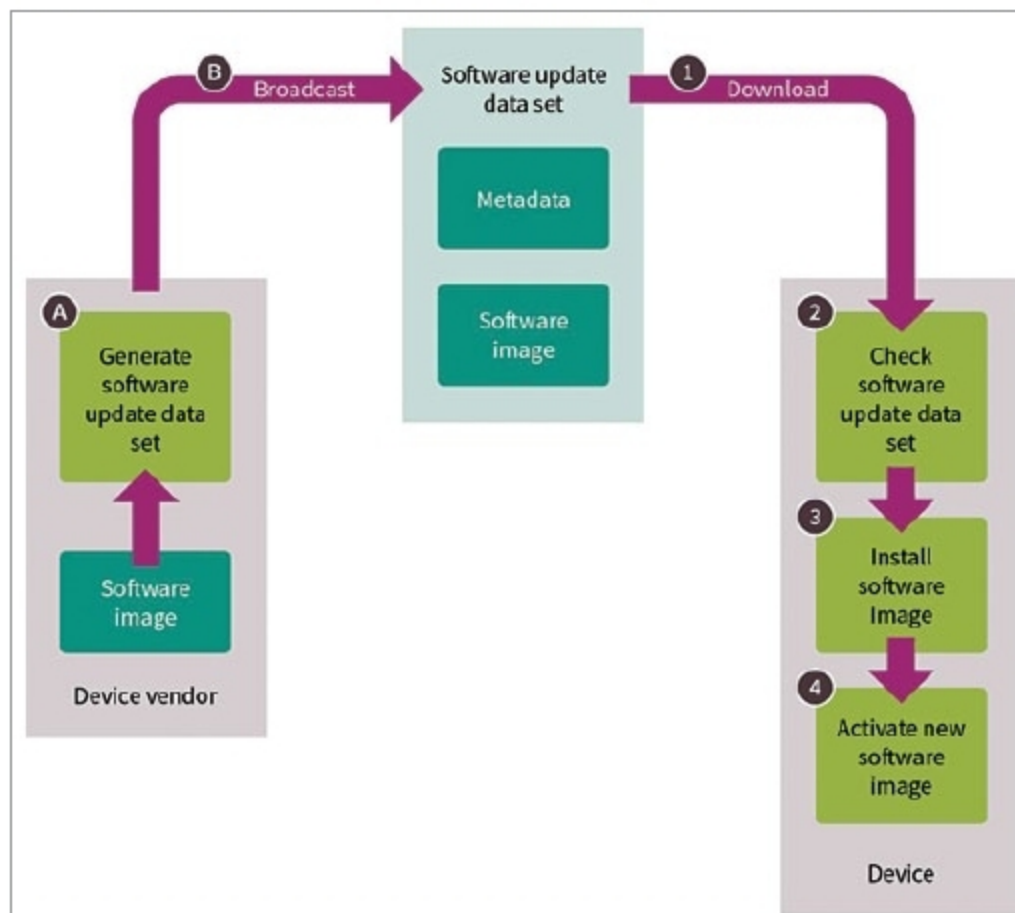


Fig. 3: Steps performed during software update (Credit: Infineon Technologies)